

MONTPELLIER MEDITERRANEE AP, France

WMO: 076430

Lat: 43.5769N

Lon: 3.9631E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	31.2	12.1	10.2	36.6	16.5	12.7	38.6	33.5	53.0	29.3	51.1	6.0	10	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	89.9	68.8	87.6	68.4	85.3	67.8	75.0	81.7	73.4	80.4	72.0	79.3	11.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	122.5	78.9	71.3	115.5	77.3	69.6	109.1	76.2	38.4	81.5	37.0	80.5	35.7	79.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.7	24.2	21.1	DB	23.7	95.4	3.3	3.3	21.3	97.8	19.3	99.7	17.5	101.6	15.1	104.0
			WB	21.5	76.8	3.7	1.5	18.9	77.8	16.7	78.7	14.7	79.5	12.0	80.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	45.7	46.7	52.3	56.9	63.6	71.2	75.7	75.0	68.2	61.7	52.5	46.8	(6)
(7)		DBStd	11.81	5.80	5.73	5.05	4.67	4.83	5.10	4.25	3.89	4.44	5.27	6.00	6.02	(7)
(8)		HDD50	485	155	116	33	4	0	0	0	0	0	2	42	133	(8)
(9)		HDD65	2922	597	511	394	246	83	7	1	1	18	124	375	565	(9)
(10)		CDD50	4054	23	25	104	211	421	637	797	775	545	366	117	33	(10)
(11)		CDD65	1017	0	0	0	3	39	194	334	311	113	23	0	0	(11)
(12)		CDH74	7885	0	0	6	25	251	1487	2914	2515	619	67	1	0	(12)
(13)		CDH80	2311	0	0	0	1	43	426	974	751	107	9	0	0	(13)
(14)	Wind	WSAvg	9.4	9.2	9.5	10.1	10.4	9.9	9.2	9.4	8.9	8.8	9.3	9.6	8.9	(14)
(15)	Precipitation	PrecAvg	34.2	3.4	3.0	3.0	2.4	1.9	1.8	0.9	1.6	4.1	4.8	3.0	3.1	(15)
(16)		PrecMax	54.1	9.6	9.8	11.0	5.8	5.0	5.8	2.4	3.9	11.5	11.7	10.2	9.6	(16)
(17)		PrecMin	19.6	0.2	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.7	0.5	0.7	0.2	(17)
(18)		PrecStd	10.9	2.7	2.8	3.0	1.8	1.4	1.6	0.6	1.3	3.1	3.3	2.4	2.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	65.9	73.5	77.1	84.8	92.6	93.2	92.3	86.7	79.8	69.9	63.8	(19)
(20)		MCWB	54.2	53.0	56.7	59.9	66.0	70.3	69.3	70.3	66.2	63.0	59.6	54.9	(20)	
(21)		2%	DB	60.2	61.6	68.1	73.2	79.8	88.2	90.1	89.0	82.8	75.3	66.0	60.6	(21)
(22)		MCWB	51.5	51.0	54.0	57.9	63.2	68.8	68.5	68.9	65.1	63.5	59.4	54.4	(22)	
(23)		5%	DB	57.1	58.7	64.8	69.9	76.4	84.8	87.8	86.5	80.2	72.6	64.1	58.6	(23)
(24)		MCWB	50.4	49.7	52.7	56.5	61.4	67.2	68.5	68.3	64.9	63.5	58.1	53.3	(24)	
(25)		10%	DB	55.1	56.3	61.7	66.5	73.5	81.6	85.5	84.3	77.4	70.4	62.2	56.5	(25)
(26)		MCWB	49.8	49.3	51.5	54.9	60.1	66.1	68.1	68.2	64.3	63.0	56.9	51.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	54.9	58.3	63.3	68.5	74.6	76.9	77.3	72.5	69.9	63.8	58.3	(27)
(28)		MCDWB	61.1	61.4	67.9	72.0	79.6	84.1	83.9	83.2	77.2	72.7	66.1	61.0	(28)	
(29)		2%	WB	53.8	53.5	56.2	60.7	66.0	72.5	74.8	75.3	71.0	68.1	61.7	56.3	(29)
(30)		MCDWB	57.7	58.1	63.5	68.8	75.3	81.5	82.0	81.4	75.7	71.1	64.6	59.2	(30)	
(31)		5%	WB	52.3	52.2	54.9	58.7	64.2	70.7	73.1	73.9	69.5	66.3	59.8	54.2	(31)
(32)		MCDWB	55.9	56.7	61.1	65.7	72.1	79.8	81.0	80.1	74.8	70.3	62.9	57.3	(32)	
(33)		10%	WB	50.6	50.6	53.7	57.2	62.7	69.0	71.6	72.5	68.0	64.6	57.9	52.3	(33)
(34)		MCDWB	54.4	55.5	59.5	63.0	69.9	77.9	79.6	79.0	74.4	68.5	61.3	55.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.8	16.5	17.3	16.5	16.8	17.4	17.4	17.4	17.1	15.1	14.9	14.6	(35)	
(36)		5% DB	MCDBR	16.8	19.7	21.8	21.4	21.4	21.0	20.5	20.6	20.6	17.9	15.7	15.5	(36)
(37)		MCWBR	11.2	12.4	12.1	11.2	10.0	9.3	8.7	9.1	9.5	9.8	10.3	10.6	(37)	
(38)		5% WB	MCDBR	13.5	16.4	17.4	17.7	17.6	17.3	16.3	15.4	15.0	12.4	11.5	12.3	(38)
(39)		MCWBR	10.2	11.4	10.9	10.8	9.9	9.1	9.3	9.3	9.5	8.7	9.1	9.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.340	0.379	0.404	0.412	0.414	0.409	0.407	0.392	0.387	0.349	0.320	(40)	
(41)		taud	2.479	2.430	2.337	2.285	2.314	2.341	2.355	2.365	2.393	2.406	2.461	2.495	(41)	
(42)		Ebn at Noon	256	270	272	273	273	272	272	269	264	249	243	246	(42)	
(43)		Edh at Noon	23	28	35	40	40	39	38	36	33	29	23	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	540	848	1245	1634	1971	2281	2280	1946	1479	917	600	471	(44)	
(45)		RadStd	60	83	99	129	132	108	94	85	86	94	73	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70
(47)	Regional (32 neighbors)	N/A	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page